

Statistics
 student marks $\Rightarrow 50, 60, 70, 80, 90$
 Number
 Central tendency

A measure of central tendency is a statistical value that represents the centre value of a dataset.

Measures central tendency
 a numeric value where the data tends to cluster.

Types: Mean, Median, Mode

Mean $\rightarrow 50, 60, 70, 80, 90$

$$\frac{50+60+70+80+90}{5} = \text{mean}$$

$$\bar{x} = \frac{\sum x_i}{n}$$

[100000, 50000, 120000, 90000, 140000]

{Even num} {Mean} {Odds}

Median $\rightarrow$ the middle value after sorting

[19, 20, 9, 18, 13]
 $\rightarrow$ [9, 13, 18, 19, 20]
 median
 $\frac{13+18}{2} = 15.5$

Mode $\rightarrow$ the most frequently occurring value

[1, 2, 3, 3, 4, 5, 6, 5, 7, 5, 8]
 $\rightarrow$ [5]

1, 2
 1+1
 2+1
 3+1
 4+1
 5+1
 $\rightarrow$ No mode

[1, 2, 3, 4, 5]
 $\rightarrow$ Mean = 3
 Median = 3
 Mode = No mode.

Class A	Class B	R
50	10	40
50	30	20
50	50	0
50	70	-20
50	90	-40

$\rightarrow$ Mean = 50
 Mean = 50

Measures of Spread

Range $\rightarrow$ Maximum - minimum

IQR $\rightarrow$ Inter quartile range

$IQR = Q_3 - Q_1$

$Q_3 = 75^{th}$ percentile

$Q_1 = 25^{th}$ percentile

{Lower bound = $Q_1 - 1.5 \times IQR$
 Upper bound = $Q_3 + 1.5 \times IQR$ }

50	10
50	30
50	50
50	70
50	90

Mean = 50

Variance $\rightarrow$ the average squared distance of each data point from the mean.

$$(\sigma)^2 = \frac{\sum (x_i - \mu)^2}{n}$$

data = [10, 20, 30, 40, 50]
 mean = 30
 $\rightarrow$ 0

[20, 10, 0, 10, 20]
 [400, 100, 0, 100, 400] $\Rightarrow \frac{1000}{5} = 200$

Standard deviation = $\sqrt{\text{variance}} = \sqrt{200} = 14.15$

Data points are typically 14.15 units away from the mean.

data = [50, 50, 50, 50, 50]
 data = [10, 30, 50, 70, 90]
 mean = 50, SD = 0
 mean = 50, SD = 14.15

marks [48, 49, 50, 51, 52] $\Rightarrow$ 50
 marks [10, 30, 50, 70, 90] $\Rightarrow$ 50
 $\Rightarrow$ [40, 20, 0, 20, 40] $\Rightarrow$ [1600, 400, 0, 400, 1600]

$n_1 \Rightarrow \frac{10}{5} = 2$
 $\sqrt{2} = 1.41$
 $\frac{4000}{5} = 800$
 $\sqrt{800} \approx 28$
 $30 \Rightarrow 900$

$M_1 = 50, SD_1 = 1.41$
 $M_0 = 50, SD_0 = 28$

10⁴
 Audit
 Stat
 A $\rightarrow$ 100
 B $\rightarrow$ 100
 C $\rightarrow$ 100
 D $\rightarrow$ 100

400 students
 4
 90/100

Z-Score Standardization

Class 1
 Mean = 50
 SD = 10
 SA = 80
 diff. from mean = $80 - 50 = 30$
 $\frac{30}{10} = 3$

Class 2
 Mean = 78
 SD = 2
 SB = 80
 diff. from mean = $80 - 78 = 2$
 $\frac{2}{2} = 1$

Z-Score $\rightarrow$ how many std. dev. a value is away from the mean.

$$Z = \frac{x - \mu}{\sigma}$$

$\rightarrow x - \text{mean}$ $\rightarrow$ how far the value is from mean.
 $80 - 50 = 30$
 student scored 30 marks above average.

$\rightarrow \frac{x - \text{mean}}{\text{std dev}} \Rightarrow \frac{30}{10} = 3$ S.D. $\rightarrow$ typical spread.

$\rightarrow$ student is 3 std dev above average.

[10, 20, 30, 40, 50] $\Rightarrow$ mean = 30
 $\Rightarrow$ std dev = 14.15

Z score = $\frac{40 - 30}{14.15} = \frac{10}{14.15} \approx 0.7$

Z score = $\frac{10 - 30}{14.15} \Rightarrow -ve$

positive $\rightarrow Z > 0 \rightarrow$ value is above mean
 negative $\rightarrow Z < 0 \rightarrow$ value is below mean
 zero $\rightarrow Z = 0 \rightarrow$ value is equal to mean.

Age = [20, 25, 30, 35, 40]

Salary = [20000, 50000, 80000, 110000, 140000]

age range $\rightarrow$ 20 to 40

Salary range $\rightarrow$ 20000 to 140000

n.k. also Salary no. are large.

$\Rightarrow$ sure standardization.

Covariance

Study hours $\propto$ Marks

measure how two variables vary together

ryhana $\leftrightarrow$ Marks
 sophia $\leftrightarrow$ marks

A numerical value

two variables

Study hours

Marks

Study hours

A	marks	study hrs
A	90	9
B	50	4
C	70	11
D	95	8

no pattern covariance ≈ 0

A	marks	study hrs
A	20	1
B	40	2
C	60	3
D	80	4

covariance positive

A	marks	study hrs
A	4	1
B	2	2
C	3	3
D	4	4

covariance -ve

Study hours $\propto$ Marks